

Practice Problem for Homework 1 Question 1

(Practice Problem for Question 1) Find the first name and last name of all passengers who have ever flown on a flight that departed from an airport located in the city of *Los Angeles* and arrived at an airport located in the city of *New York*. Cities may have multiple airports. Return each passenger only once, even if they took multiple such flights.

Answer:

Step 1: Identify Los Angeles airports.

$$LA = \sigma_{\text{City}='Los Angeles'}(\text{Airports})$$

Step 2: Identify New York airports.

$$NY = \sigma_{\text{City}='New York'}(\text{Airports})$$

Step 3: Find all flights departing from Los Angeles airports.

$$FromLA = \text{Flights} \bowtie_{\text{DepartureAirportCode}=\text{Code}} LA$$

Step 4: From those, keep only flights arriving at New York airports.

$$LAtoNY = FromLA \bowtie_{\text{ArrivalAirportCode}=\text{Code}} NY$$

Step 5: Retrieve the passengers who flew on those flights. We first connect flights to tickets, and then tickets to passengers:

$$Pax = \text{Passengers} \bowtie \text{Tickets} \bowtie \text{TicketFlights} \bowtie_{\text{TicketFlights.FId}=LAtoNY.FId} LAtoNY$$

Step 6: Project the required attributes and remove duplicates.

$$\pi_{\text{FirstName,LastName}}(Pax)$$

Final Answer

$$\pi_{\text{FirstName,LastName}} \left(\text{Passengers} \bowtie \text{Tickets} \bowtie \text{TicketFlights} \bowtie_{\text{TicketFlights.FId}=LAtoNY.FId} \left(\left(\text{Flights} \bowtie_{\text{DepartureAirportCode}=\text{Code}} \sigma_{\text{City}='Los Angeles'}(\text{Airports}) \right) \bowtie_{\text{ArrivalAirportCode}=\text{Code}} \sigma_{\text{City}='New York'}(\text{Airports}) \right) \right)$$

This expression returns the first and last names of all distinct passengers who have flown on at least one flight departing from Los Angeles and arriving in New York.